Ion CSA v3 response study

Generated production results

November 6, 2026

1 Matched production and timing

This production contains 8,000 bb0nu events at 3 mm pitch and 30 ms frames. The shared realization places ion drift uniformly between 5 and 145 cm, with complementary electron drift in the 150 cm chamber. Every event ID, primary-pair checksum and drift distance was checked against the existing matched common realization and 4 mm ion file. The electron readout was preserved. All events and signed response buckets are retained. The ion ledger is 897,306,359, exactly equal to the primary-pair total. The CSA used 15,178 nonlinear histories and 6 SPICE fallback histories among 1,118,323 pad histories. Production creates no SPICE control histories. Fresh filtered outputs were asserted before the run; actual circuit-deck, model, kernel-waveform and calibration checksums are saved in preflight.toml. Julia sendfile reported an error after producing a byte-identical publication copy. The saved copy was verified by SHA256, its production metadata completed and its event ledgers validated before registration. Numerical timing covers the completed fresh calculation; recovery and registration are timed separately.

Stage	Seconds
Ion diffusion and collection	39.4565
Covariance calibration setup	45.9391
Linear predictions and current preparation	90.9843
Nonlinear response and storage	61.7863
SPICE fallback workflow	227.6810
Three-frame reconstruction and summary	145.9292
Figures and presentation	0.9305
Event work excluding calibration and final presentation	526.4824
Event work per event	0.0658
Publication recovery and registration	138.1008

The numerical timer includes screening, nonlinear response, noise, reconstruction, output writing and consistency checks. Covariance calibration and final presentation are separated. Event work includes fallback orchestration and its internal report generation when SPICE fallbacks occur. Stage and complete times are nested; the per-event value divides measured event work by 8,000. The study processes 8,000 event IDs per sample at 3 mm pitch and 30 ms frame width. Every pixel with charge retains its complete event history, including empty frames, two quiet leading frames and a three-second recovery window. There is no S/N cut. Signed reconstructed counts, including negative noise fluctuations, are retained.

The circuit is the provisional v3 design, nominal 20 s discharge, with 5 ms centred voltage averages, 1.25 pF external capacitance and 3 fF feedback. The input uses uniform mean collection within each frame and the established central 0.5 mm square-pad weighting benchmark in an infinite grounded plane. The pad weighting potential is one, with zero on the surrounding plane and at infinity, in homogeneous dielectric. The ion speed is 3 m/s through the finite 5 mm gap. The charge integral is $e[1 - \phi_w(g)]$ per collected ion.

Sample	ENC before	ENC after	Bias before (%)	Bias after (%)
Double beta	54.63	55.32	+4.151	-0.205

The same transistor circuit was recalibrated at 20, 30 and 40 ms. Noise uses its one-sided stationary operating-point spectrum from 0.0001 Hz to 1 MHz, filtered by the 5 ms average and the decay-corrected frame difference. Gaussian samples retain temporal covariance; pad channels are independent. This stationary noise approximation is also applied to compressed mean signals.

Frame (ms)	Fitted τ (s)	Filtered gain (μV/ion)	ENC (ions RMS)	Lag-1 correlation
20	19.5441	40.157	51.26	-0.3292
30	19.5440	41.042	55.32	-0.2975
40	19.5440	41.479	58.95	-0.2738

S/N is the noiseless recovered signal divided by frame ENC, evaluated over all positive input pixel–frame cells. The cell-weighted and event-weighted means are shown separately. Compression compares the noiseless transistor response with the calibrated linear prediction; the flags require a deficit of at least one frame ENC. A pad history is counted once per event if any positive-input bucket passes the indicated local-loss criterion.

Sample	Events	ENC	Mean S/N	Event mean S/N	Loss50 events (%)	Deficit (%)	Total bias (%)
Double beta	8000	55.32	3.181	3.903	1.65	0.174	-0.205

The positive-cell compression deficit is $\sum_{N_n>0} \max(\widehat{N}_{\mathrm{linear,n}} - \widehat{N}_{\mathrm{CSA,n}}, 0)/\sum N_n$. The signed total bias is $(\sum \widehat{N}_{\mathrm{CSA,n}} - \sum N_n)/\sum N_n$, including quiet frames and recovery. The linear prediction's own calibration bias is saved separately, as is the noisy summed estimate. The corrected reconstruction solves $M\widehat{\mathbf{N}} = \mathbf{y}$ using the saved three-frame response. The same banded solve is applied to the noiseless signal and the existing noise realisation. Its covariance is $M^{-1}C_y M^{-T}$. Quiet boundary frames are retained, and exact history-dependent frame variances enter S/N. This removes the linear normalization bias and temporal mixing while preserving the circuit compression.

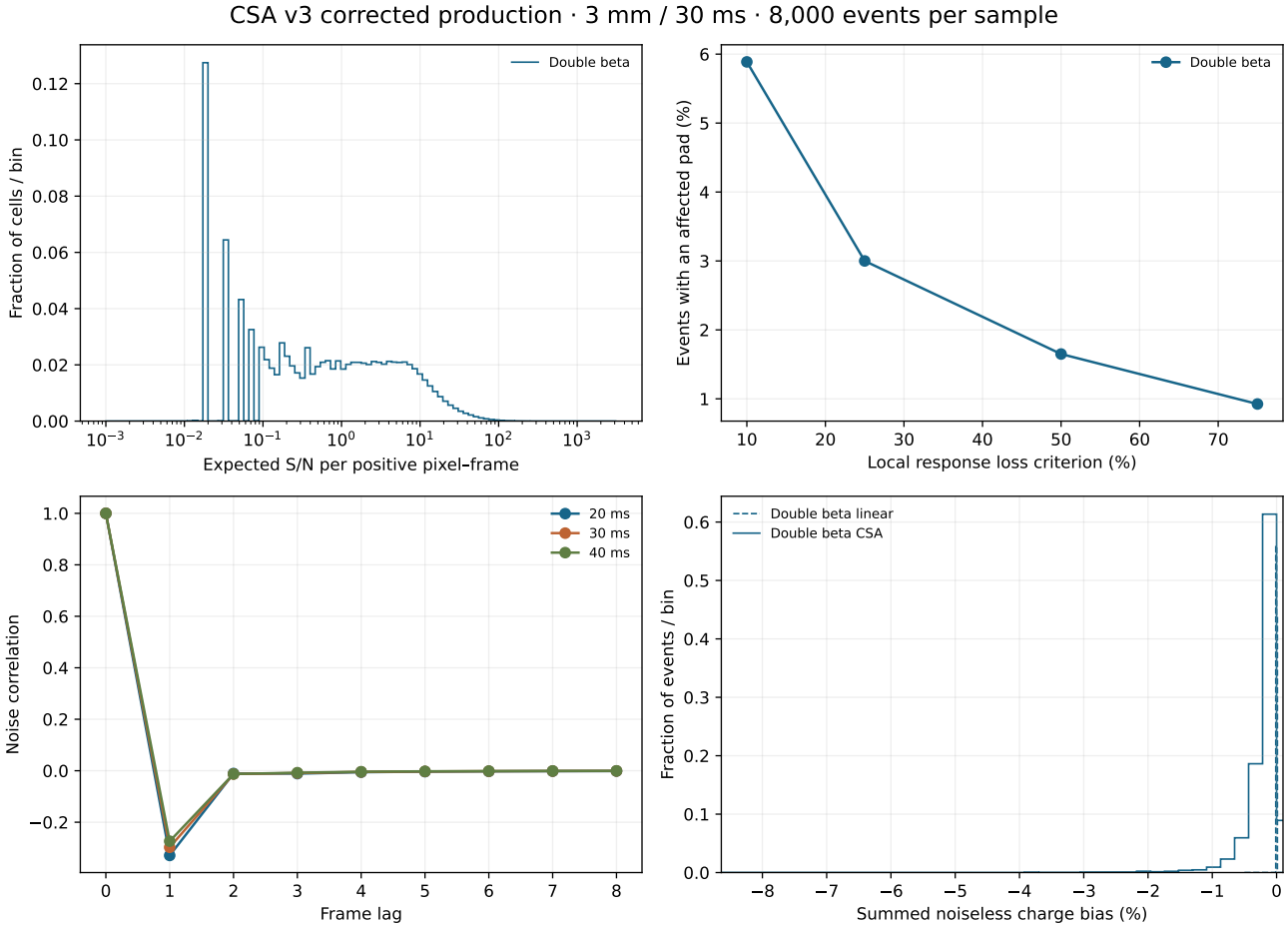


Figure 1: Saved study quality, compression, covariance and charge-bias diagnostics. Dashed charge-bias curves use the linear estimator; solid curves include transistor response where directly simulated.

Production mode uses the saved, hash-checked nonlinear calibration and voltage guard. SPICE is used for histories outside the calibrated domain. No new SPICE control or solver-refinement histories are generated. Independent checks cover finite-path charge, spectral integration, covariance convergence and positive definiteness, sampled zero-input mean/variance/covariance, input ion ledgers and Float32 storage errors relative to Float64 calculations.

Response files use `nausicaa.ion_csa_response.pilot.v2`. Each saves the selected IDs, parent file/hash, exact cards, code identity, stable per-pad seed, frame covariance, input counts, mean voltages, linear/noiseless/noisy recovered ions and compression flags. Files are written atomically and marked complete after the requested event ledgers pass.

The companion `ion_csa.html` provides sample and compression-severity selectors, diagnostic distributions and typical/weak/compressed pad histories. S/N is a quality observable; final analysis determines acceptable performance. Spatial noise occupancy requires a finite readout area; the present controls use zero-input histories on the selected channels.

From the repository root:

```
work/event-viewer-venv/bin/python scripts/production/run_ion_csa_correction.py
# Presentation only:
work/event-viewer-venv/bin/python scripts/production/present_ion_csa_pilot.py \
  --results RESULTS_DIRECTORY
```